

# Data Quality and Validation in ETL

## Question 1

**Define Data Quality in the context of ETL pipelines. Why is it more than just data cleaning?**

**Answer:**

Data Quality refers to the accuracy, completeness, consistency, validity, and reliability of data in an ETL pipeline. It is more than data cleaning because it not only removes errors but also ensures that data follows business rules, maintains consistency across systems, and remains trustworthy for analysis and reporting.

## Question 2

**Explain why poor data quality leads to misleading dashboards and incorrect decisions.**

**Answer:**

Poor data quality can result in inaccurate reports, duplicate records, missing values, and inconsistent information. These issues create misleading dashboards and incorrect KPIs, causing businesses to make wrong decisions based on unreliable data.

## Question 3

**What is duplicate data? Explain three causes in ETL pipelines.**

**Answer:**

Duplicate data occurs when the same record appears multiple times in a dataset.

**Causes:**

1. Multiple data sources loading the same record.
2. ETL jobs running more than once.
3. Missing primary key or duplicate detection rules.

**Differentiate between exact, partial, and fuzzy duplicates.**

| Type            | Description                      | Example                       |
|-----------------|----------------------------------|-------------------------------|
| Exact Duplicate | Records are completely identical | Same customer record repeated |

|                   |                                     |                                           |
|-------------------|-------------------------------------|-------------------------------------------|
| Partial Duplicate | Some fields match but others differ | Same customer with different phone number |
| Fuzzy Duplicate   | Similar but not exactly same        | "Rahul Mehta" and "Rahul M."              |

#### Question 5

**Why should data validation be performed during transformation rather than after loading?**

**Answer:**

Data validation during transformation helps identify and correct errors before data reaches the target system. This reduces rework, improves performance, and prevents incorrect data from being stored in the database.

#### Question 6

**Explain how business rules help in validating data accuracy. Give an example.**

**Answer:**

Business rules define acceptable values and conditions for data. They ensure that only valid data enters the system.

**Example:**

Transaction Amount must be greater than zero. If a record contains a negative amount, it will be rejected during validation.

#### Question 7

SELECT

Customer\_ID,

Product\_ID,

Txn\_Date,

Txn\_Amount,

COUNT(\*) AS Duplicate\_Count

FROM Sales\_Transactions

GROUP BY

Customer\_ID,

Product\_ID,

Txn\_Date,

Txn\_Amount

HAVING COUNT(\*) > 1;

**Answer:**

Duplicate transaction found for Customer\_ID C101 and Product\_ID P11 with count 3.

**Question 8**

SELECT DISTINCT s.Customer\_ID

FROM Sales\_Transactions s

LEFT JOIN Customers\_Master c

ON s.Customer\_ID = c.CustomerID

WHERE c.CustomerID IS NULL;

**Answer:**

Customer\_IDs C105 and C106 violate referential integrity because they do not exist in Customers\_Master.